Restoring electron spin and the quantum vacuum in variational lattice QED simulations of strong-field plasmas

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Abstract

Real-time lattice scalar QED, obtained by extremising an exactly gauge-invariant discrete action and solved in the classical-statistical regime, has been proposed as a first-principles tool for plasmas in fields approaching the critical field $E_c = m^2/q$. We determine which relativistic-quantum plasma physics this tree-level scheme reproduces and which it cannot, after reproducing its published linear-wave and laser–plasma demonstrations in an independent implementation (the pair yields of the latter are not yet converged at the published resolution). The scheme reproduces superradiant Klein reflection from a supercritical Sauter step to 9.8×10^{-6} of the exact result, with gauge-invariant observables identical to round-off (7.6×10^{-12}) in three gauges whose link phases wind repeatedly around the $U(1)$ circle, and it reproduces the complex frequencies of Landau-damped Langmuir waves in a warm plasma with quantum recoil to within 0.24%, where classical and semiclassical kinetic theories are off by factors up to 5.5. Two premises of the scheme fail for fundamental reasons, at every lattice resolution. First, spin-zero matter misplaces the cyclotron line of a magnetised plasma at $B = B_c$ by 20.2% and the rest energy of the lowest Landau level by 41.4%; a gauge-invariant lattice Pauli term restores both to within 0.19% and 0.08%. Second, the classical vacuum produces no spontaneous pairs at all, and classical seed noise produces pairs only in proportion to the seed; a half-quantum Bogoliubov extension of the discrete action restores the exact Sauter-pulse pair spectrum (density to 7.6×10^{-7}) and the decay of a supercritical field into vacuum-pair plasma oscillations. Both extensions preserve exact local charge conservation. Tree-level lattice QED is thus exact for stimulated relativistic-quantum processes but blind to the spin magnetic moment and to the zero-point fluctuations that seed spontaneous ones.

1 Introduction

Plasmas exposed to fields approaching the critical field $E_c = m^2/q$ (the Sauter–Schwinger field, 1.3×10^{18} V/m for electrons) or to magnetic fields approaching $B_c = m^2/q$ (4.4×10^9 T) are no longer described by the kinetic theory of point particles. Pairs are created out of the vacuum or out of the plasma itself [1, 2, 3, 4, 5], wave dynamics reaches the Compton scale, Landau quantisation reorganises the response of magnetised matter [6, 7, 8, 9], and the energy and momentum carried away by created pairs feed back on the fields that create them [10, 11, 12]. Such conditions are encountered in the focus of multi-petawatt lasers [13, 14, 15, 16], in pulsar and magnetar magnetospheres [8, 9], and in proposals for gamma-ray sources based on dense pair plasmas [17, 18]. The standard numerical response—particle-in-cell codes augmented with Monte-Carlo source terms for pair creation and photon emission [15, 14, 19]—treats every strong-field process as an independent, local and incoherent event, and therefore cannot capture interference between coexisting processes [20], stimulated (Bose-enhanced) channels, or the exact accounting of energy and momentum once many pairs are present.

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A field-theoretic alternative is to solve quantum electrodynamics on a space-time lattice in real time. When occupation numbers are large and the coupling is weak, tree-level contributions dominate and the quantum dynamics can be approximated by classical field equations averaged over an ensemble of initial conditions—the classical-statistical approximation [21, 22, 23, 24, 25]. Shi, Xiao, Qin and Fisch [26] turned this idea into a plasma-physics tool: they discretised the action of scalar QED with Wilson lines for the matter coupling and a Yee-type field-strength term for the gauge field [27, 28], obtained explicit equations of motion by extremising the discrete action, and showed that the resulting scheme conserves charge exactly, satisfies the Bianchi identities by construction, reproduces cold-plasma dispersion relations, and exhibits a transition from wakefield acceleration to pair production when an intense gamma-ray laser drives a dense bosonic plasma. Dirac-field versions with canonical symplectic integrators have since been developed [29], and the approach has been reviewed in the broader context of strong-field plasma theory [30, 31].

The scheme sits at the intersection of several lines of work. Its discrete action combines Wilson's compact link variables [32, 33, 34] with a Yee-type non-compact field term [27], in the spirit of the gauge-invariant Maxwell–Klein–Gordon discretisations of numerical analysis [28, 35] and of the structure-preserving particle-in-cell algorithms of plasma physics [36, 37, 38]. Its use of classical field equations with an ensemble of initial conditions is the classical-statistical (truncated Wigner) approximation of non-equilibrium quantum field theory [39, 21, 40, 24, 41, 42], which has been used with fermionic mode functions to simulate the Schwinger effect and string breaking in $1+1$ dimensions [43, 44, 45, 46, 47, 48]. Pair creation in strong fields is otherwise studied with quantum kinetic equations and mode functions in prescribed or mean fields [49, 50, 11, 51, 52, 53, 54], and Klein tunnelling and superradiance at potential steps have a long history of their own [55, 56, 57, 58, 59, 60, 61, 62, 63, 64]. On the plasma side, relativistic and quantum dispersion theory [65, 66, 67, 68, 69, 70, 71, 72, 73, 74] and spin-resolved quantum plasma models [75, 76] provide the analytic baselines against which a lattice field theory of plasmas must be judged. Finally, because the scheme is local and unitary at the level of its building blocks, it is a natural target for quantum simulation of lattice gauge theories [77, 78, 79]. To our knowledge no systematic test of the tree-level lattice scheme against exact relativistic-quantum results beyond the original demonstrations has been reported, and its behaviour in magnetised and vacuum-dominated regimes has not been examined.

For such a scheme to become a quantitative instrument, one must know precisely which physics it reproduces and which it cannot, and why. Two structural features control the answer. First, *exact discrete gauge invariance*: the matter field couples to the gauge field only through compact $U(1)$ link variables, which is what makes charge conservation exact but also raises the question of whether sustained supercritical fields—whose vector potential grows without bound in the temporal gauge [26]—can be represented at all. Second, *the tree-level, spin-zero premise*: the matter is a classical scalar field and the vacuum is empty. Both premises are harmless for classical plasma physics, which ignores spin and quantum fluctuations, but neither is obviously harmless in the regimes that motivate the method.

In this paper we establish the following. (i) The scheme reproduces *superradiant Klein reflection* [55, 56, 59, 58] of a charged-boson wave packet from a supercritical, smooth electrostatic step to a relative accuracy of 9.8×10^{-6} at the finest resolution, in three different gauges whose link variables wind many times around the $U(1)$ circle, with gauge-invariant observables agreeing to round-off. A bounded static potential drop is therefore represented exactly, however large the vector potential. (ii) Its multi-species formulation reproduces the complex frequencies of *Landau-damped Langmuir waves in a warm, mildly relativistic plasma in the quantum-recoil regime*, where classical, non-relativistic and semiclassical kinetic theories mispredict the damping rate by factors of up to 5.5; we derive the tree-level warm-plasma susceptibility for the lattice model and show that the simulation selects it. (iii) In a magnetar-strength field $B = B_c$, the spin-zero scheme misplaces the cyclotron line of a dilute plasma by 20.2% and the rest energy of the lowest Landau level by 41.4% relative to the spin-$\frac{1}{2}$ values, at every lattice resolution: the premise that particle spin is irrelevant to plasma dynamics fails when $\hbar\omega_c \sim mc^2$. We construct a gauge-invariant *lattice Pauli coupling* that restores the Dirac Landau spectrum while preserving exact charge conservation. (iv) The scheme creates no pairs at all from the vacuum, however strong the field and however fine the lattice, and classical seed fluctuations produce only stimulated pairs in proportion to the arbitrary seed: spontaneous pair creation is absent at tree level. We construct a *half-quantum Bogoliubov extension* of the discrete action that restores the exact Sauter-pulse pair spectrum and, when the gauge field is dynamical,

the decay of a supercritical field into vacuum-pair plasma oscillations, again with exact charge conservation. Both demonstration problems of Ref. [26] are reproduced in the same implementation, which rules out implementation errors as the origin of (iii) and (iv).

Taken together, the results draw a sharp line: tree-level lattice QED is quantitatively exact for *stimulated* relativistic-quantum processes—Klein superradiance, quantum Landau damping, pair creation seeded by an existing plasma—but is blind to the two ingredients that distinguish electrons in the vacuum from classical waves, namely the spin magnetic moment and the zero-point fluctuations that seed *spontaneous* processes. Both can be restored within the same variational, exactly gauge-invariant framework.

2 Methodology

We use natural units $\hbar = c = \varepsilon_0 = 1$, the metric signature $(+,-,-,-)$ and lower-index link variables A_μ, so that $A_0 = A^0$ is the scalar potential and $A_l = -A^l$. Throughout, $q = 0.3$ ($q^2/4\pi \simeq 1/137$) and the boson mass is $m = 1$.

2.1 Variational lattice scalar QED in one dimension

Fields and Wilson lines. As in the one-dimensional simulations of Ref. [26], space is discretised along x with spacing Δx and periodic boundaries, while each transverse direction consists of a single periodic cell of size $\Delta_\perp$ (we take $\Delta_\perp = \Delta x$ unless stated otherwise). Complex matter fields $\phi^n_{f,i}$ live on vertices (t_n, x_i); the index f labels charged species, which in the original scheme is a single field. A species may carry a fixed transverse momentum p_f along y (Sec. 2.2.3). The gauge 1-form lives on edges: $A^{n+1/2}_{0,i}$ on time-like edges, $A^n_{1,i+1/2}$ on x-edges and $A^n_{2,i}$, $A^n_{3,i}$ on the transverse edges attached to vertex i. Pull-back covariant differences are built from Wilson lines,

$$(D_0\phi)^{n+1/2}_i = \tfrac{1}{\Delta t}\big(e^{-iq\Delta t A^{n+1/2}_{0,i}}\phi^{n+1}_i - \phi^n_i\big), \quad (D_1\phi)^n_{i+1/2} = \tfrac{1}{\Delta x}\big(e^{-iq\Delta x A^n_{1,i+1/2}}\phi^n_{i+1} - \phi^n_i\big),$$
$$(D_2\phi)^n_i = \tfrac{1}{\Delta_\perp}\phi^n_i\big(e^{i\theta^n_{2,i}} - 1\big), \qquad \theta^n_{2,i} = (p_f - qA^n_{2,i})\Delta_\perp, \tag{1}$$

and similarly for D_3. Field strengths are exterior derivatives of the links: $E^{n+1/2}_{x,i+1/2} = (A^{n+1}_1 - A^n_1)_{i+1/2}/\Delta t - (A^{n+1/2}_{0,i+1} - A^{n+1/2}_{0,i})/\Delta x$, $E^{n+1/2}_{y,i} = (A^{n+1}_{2,i} - A^n_{2,i})/\Delta t$, $B^n_{z,i+1/2} = -(A^n_{2,i+1} - A^n_{2,i})/\Delta x$ and $B^n_{y,i+1/2} = (A^n_{3,i+1} - A^n_{3,i})/\Delta x$, so that Faraday's law and $\nabla\cdot\mathbf{B} = 0$ hold identically.

Discrete action and equations of motion. The discrete action is the Yee-type action of Ref. [26] [their Eqs. (14)–(15)], summed over species,

$$S_d = \Delta t\,\Delta x\,\Delta_\perp^2 \sum_{n,i}\Big\{\sum_f w_f\Big[|D_0\phi_f|^2 - \sum_{l=1}^{3}|D_l\phi_f|^2 - m_f^2|\phi_f|^2\Big] + \tfrac{1}{2}\big(\mathbf{E}^2 - \mathbf{B}^2\big) - \rho_{\rm ext}A_0\Big\}, \tag{2}$$

where $w_f = 1$ for physical species and $\rho_{\rm ext}$ is a static external charge density (immobile ions or the electrodes that create a potential step). Variation with respect to $\bar\phi^n_{f,i}$ gives the discrete Klein–Gordon equation [Eq. (16) of Ref. [26]] in explicit form,

$$\phi^{n+1}_i = e^{iq\Delta t A^{n+1/2}_{0,i}}\Big[\big(2 - 2C^2 - \Delta t^2 m^2 + \Delta t^2 T_{\perp,i}\big)\phi^n_i + C^2\big(e^{-iq\Delta x A_{1,i+1/2}}\phi^n_{i+1} + e^{iq\Delta x A_{1,i-1/2}}\phi^n_{i-1}\big) - e^{iq\Delta t A^{n-1/2}_{0,i}}\phi^{n-1}_i\Big], \tag{3}$$

with $C = \Delta t/\Delta x < 1$ and $T_{\perp,i} = \sum_{l=2,3} 2(\cos\theta_{l,i} - 1)/\Delta_\perp^2$, which reduces to $-(p_f - qA_2)^2 - q^2A_3^2$ when $\theta_l \to 0$. Variation with respect to A_0, A_1 and $A_{2,3}$ gives the discrete Gauss law, $(E_{x,i+1/2} - E_{x,i-1/2})/\Delta x = J^{n+1/2}_{0,i} + \rho_{{\rm ext},i}$, and the discrete Ampère laws

$$\frac{E^{n+1/2}_{x,i+1/2} - E^{n-1/2}_{x,i+1/2}}{\Delta t} = J^n_{1,i+1/2}, \quad \frac{E^{n+1/2}_{y,i} - E^{n-1/2}_{y,i}}{\Delta t} = \frac{A^n_{2,i+1} - 2A^n_{2,i} + A^n_{2,i-1}}{\Delta x^2} + J^n_{2,i}, \tag{4}$$

with the gauge-invariant charge and current densities

$$J_0 = \sum_f \frac{2w_f q}{\Delta t}\,\mathrm{Im}\big(e^{-iq\Delta t A_0}\phi_f^{n+1}\bar{\phi}_f^{n}\big), \qquad J_1 = \sum_f \frac{2w_f q}{\Delta x}\,\mathrm{Im}\big(e^{-iq\Delta x A_1}\phi_{f,i+1}\bar{\phi}_{f,i}\big),$$

$$J_2 = \sum_f \frac{2w_f q}{\Delta_\perp}|\phi_{f,i}|^2 \sin\theta_{2,i}. \tag{5}$$

These are Eqs. (17)–(20) of Ref. [26]; positive-frequency quanta carry charge $-q$. The gauge is fixed either by the temporal condition $A_0 = 0$ or by the discrete Lorenz condition $A_{0,i}^{n+1/2} = A_{0,i}^{n-1/2} + C(A_{1,i+1/2}^{n} - A_{1,i-1/2}^{n})$ [Eq. (B10) of Ref. [26]]. Initial data $\phi^0, \phi^1, A^0, A_0^{1/2}$ are completed by solving the Gauss law for A_1^1; thereafter the scheme is fully explicit (Fig. 3 of Ref. [26]). The code (`proposal/code/lsqed/core.py`) implements exactly Eqs. (3)–(5) with Numba kernels; no other discretisation is used anywhere in this work.

Exact and approximate invariants. Invariance of S_d under $\phi_f \to e^{iq\alpha}\phi_f$, $A_\mu \to A_\mu + \delta_\mu\alpha$ implies the local conservation law $(J_0^{n+1/2} - J_0^{n-1/2})/\Delta t = (J_{1,i+1/2} - J_{1,i-1/2})/\Delta x$, which makes the total charge $Q^{n+1/2} = \sum_i J_{0,i}\Delta x$ constant to round-off and preserves the Gauss law. The discrete energy

$$\mathcal{U}^{n+1/2} = \sum_i \Delta x\Big\{\sum_f w_f\Big[|D_0\phi_f|^2 + \mathrm{Re}\,e^{iq\Delta t A_0}\big(m^2\phi_f^n\bar{\phi}_f^{n+1} + \textstyle\sum_l (D_l\phi_f)^n\overline{(D_l\phi_f)^{n+1}}\big)\Big] + \tfrac{1}{2}\mathbf{E}^2 + \tfrac{1}{2}\mathbf{B}^n\cdot\mathbf{B}^{n+1}\Big\} \tag{6}$$

[Eqs. (A18)–(A20) of Ref. [26], per unit transverse area] is gauge invariant and conserved up to $\mathcal{O}(q\Delta^2)$ errors; it is exactly conserved when $q \to 0$. We monitor Q, $\mathcal{U}$, the matter and field contributions $\mathcal{U}_\phi, \mathcal{U}_A$, and the Gauss residual in every run.

2.2 Physical settings and pre-stated predictions

Each setting below probes a distinct regime, has a sharp prediction derived from first principles, a pre-stated tolerance (all collected in `logs/tolerances.json`, written after small pilot runs and before the production runs of these four settings), and a *primary diagnostic*—the class of implementation error it is most sensitive to.

2.2.1 Superradiant reflection from a supercritical Sauter step (primary diagnostic: gauge invariance)

Static external charges create the potential energy $U(x) = \frac{V_0}{2}[1 + \tanh(x/\ell)]$ for positive-frequency quanta ($U = -qA_0$), with $V_0 = 4m$ and $\ell = 1/m$, i.e. a peak field $qE = V_0/2\ell = 2m^2 = 2E_c$. A mode $e^{-i\varepsilon t}\phi(x)$ obeys $\phi'' + [(\varepsilon - U)^2 - m^2]\phi = 0$. In the Klein zone $\varepsilon - V_0 < -m$ the transmitted wave has negative kinetic energy (it is an antiparticle) and must carry positive group velocity, $\phi \propto e^{-i\kappa x}$ with $\kappa = [(\varepsilon - V_0)^2 - m^2]^{1/2}$. With $z = [1 + \tanh(x/\ell)]/2$ the mode equation becomes $z(1-z)\phi_{zz} + (1-2z)\phi_z + [a^2/z + b^2/(1-z) - c^2]\phi = 0$ with $a = k\ell/2$, $k = (\varepsilon^2 - m^2)^{1/2}$, $b = \kappa\ell/2$, $c = V_0\ell/2$; its solutions are hypergeometric, and the connection formula between $z = 1$ (transmitted) and $z = 0$ (incident and reflected) together with $|\Gamma(\frac{1}{2} + iy)|^2 = \pi/\cosh\pi y$ gives the reflected-to-incident charge-flux ratio

$$R = \frac{\cosh[\pi\ell(k+\kappa)] + \cosh\nu}{\cosh[\pi\ell(\kappa-k)] + \cosh\nu} > 1, \qquad R - 1 = \frac{2\sinh(\pi\ell k)\sinh(\pi\ell\kappa)}{\cosh[\pi\ell(\kappa-k)] + \cosh\nu}, \tag{7}$$

with $\nu = \pi(V_0^2\ell^2 - 1)^{1/2}$. $R - 1$ is the mean number of pairs created per incident boson [59, 57, 63, 64]: bosonic Klein reflection is *stimulated* pair creation, so it is a tree-level effect. We verified Eq. (7) against an independent high-precision integration of the continuum mode equation (agreement to 10^{-10}, `logs/theory_sauter_checks.json`). The incident packet is a small-amplitude superposition of exact lattice positive-frequency plane waves (spatial width $10/m$); the prediction is R averaged over its charge spectrum. The test energies are $\varepsilon/m = 1.2, 1.5, 2.0, 2.5$ (Klein zone; $R = 1.15$–1.28) and $\varepsilon/m = 5.5$ (normal zone, $R \approx 10^{-3}$). *Tolerance:* $|R_{\mathrm{sim}}/R - 1| \le 10^{-3}$

for all Klein-zone energies at every resolution (pilot discretisation error $\leq 4 \times 10^{-4}$ at the coarsest grid; tail leakage $< 10^{-6}$). The exponentially small normal-zone reflection is reported as a convergence trend only.

The same physical state is evolved in three gauges: (a) a "static" gauge (discrete Lorenz condition with the time-independent $A_0 = -U/q$), (b) the temporal gauge, in which A_1 grows linearly in time inside the step so that the Wilson phases $q\Delta x A_1$ wind repeatedly around the circle, and (c) the temporal gauge after a random, time-independent gauge transformation with $|q\alpha|$ up to 15. Because Eqs. (3)–(5) are exactly covariant, gauge-invariant observables (J_0, E_x, $|\phi|^2$, R) must agree to round-off. *Gauge criterion:* maximum relative difference $\leq 10^{-10}$. An error in any Wilson-line phase or in the Lorenz update would violate this at order unity.

2.2.2 Landau damping in a warm relativistic plasma with quantum recoil (primary diagnostic: energy conservation)

A warm plasma is represented, as allowed by the species sum in Eq. (2), by one charged species per momentum mode: $\phi_j = c_j e^{ip_j x - i\omega_j t}$, $p_j = 2\pi j/L$, with $|c_j|^2 = n_0 g_j/2E_j$, $g_j \propto e^{-p_j^2/2\sigma_p^2}$, $\sum_j g_j = 1$ (plus immobile ions of density n_0). To derive the tree-level response, perturb one mode $\phi_0 = ce^{ipx-iEt}$ by a longitudinal potential $A_0 = ae^{i(kx-\omega t)} + \text{c.c.}$ Linearising $(D_0^2 - \partial_x^2 + m^2)\phi = 0$ gives sidebands $\delta\phi_\pm = qa_\pm(2E \pm \omega)\phi_0 e^{\pm i(kx-\omega t)}/D_\pm$, $D_\pm = E_{p\pm k}^2 - (E \pm \omega)^2$, and inserting them into the charge density $j^0 = (q/i)(\bar{\phi} D_0 \phi - \text{c.c.})$ (including the seagull term $-2q^2 A_0|\phi|^2$) and the Gauss law yields $\epsilon = 1 + \chi$ with

$$\chi(k,\omega) = \frac{q^2 n_0}{k^2} \sum_j \frac{g_j}{2E_j} \left[\frac{(2E_j + \omega)^2}{E_{p_j+k}^2 - (E_j + \omega)^2} + \frac{(2E_j - \omega)^2}{E_{p_j-k}^2 - (E_j - \omega)^2} + 2 \right]. \tag{8}$$

For $g_j = \delta_{j0}$ this reduces exactly to the cold susceptibility $\chi_p = \omega_p^2(k^2 - \omega^2 + 4m^2)/[(\omega^2 - k^2)^2 - 4m^2\omega^2]$ of Ref. [26] [their Eq. (23)]. Its resonant part, $\chi_{\text{res}} = (q^2 n_0/k^2)\sum_j W_j[g(p_j + k) - g(p_j)]/(\omega - E_{p_j+k} + E_j)$ with $W_j = (E_j + E_{p_j+k})^2/4E_j E_{p_j+k}$, is a relativistic Lindhard function [65]: Landau damping [80] is controlled by the *finite difference* $g(p+k) - g(p)$ (quantum recoil) at the relativistic resonance $\omega = E_{p+k} - E_p$. For a continuous Gaussian distribution we continue χ to $\text{Im}\,\omega < 0$ by integrating over the shifted contour $p = s - i\eta$, $\eta = 1.5\sigma_p$, which passes below the resonant poles; the damped root $\omega = \omega_r - i\gamma$ of $\epsilon = 0$ is found by Newton iteration and is independent of η to 10^{-16} (`theory/landau.py`, `logs/predictions.json`). The same contour method reproduces the classical Vlasov root obtained from the plasma dispersion function to all printed digits.

We take $\omega_p = 0.2m$, $\sigma_p = 0.25m$ and $k = 0.3125$–$0.5625\,m$, i.e. $k\lambda_D = 0.39$–0.70 with $\lambda_D = \sigma_p/m\omega_p$, recoil $k/2m$ comparable to the thermal velocity, and resonant momenta $p \approx 0.6m$ (Lorentz factor 1.16). In this regime the relativistic-quantum prediction differs from the classical Vlasov, non-relativistic quantum (Lindhard) and relativistic classical predictions by factors between 0.79 and 5.5 in γ (Table 2), so the test is discriminating. A coherent wave is launched by the phase kick $\phi_j \to \phi_j e^{i\epsilon\sum_k \sin(kx+\vartheta_k)}$, $\epsilon = 2 \times 10^{-3}$, which perturbs velocities but not the charge density and hence leaves the Gauss law intact; runs with $+\epsilon$, $-\epsilon$ and 0 are made, and the odd part $\delta E_k = (E_k^+ - E_k^-)/2$ is fitted by a matrix-pencil decomposition [81] refined by nonlinear least squares in the window $40 \leq mt \leq 300$. *Tolerances:* $|\omega_r^{\text{sim}}/\omega_r - 1| \leq 0.5\%$ for all k and $|\gamma^{\text{sim}}/\gamma - 1| \leq 5\%$ for the modes with $\gamma/\omega_r \geq 0.02$ (the least-damped mode, $\gamma/\omega_r = 0.003$, decays by only 15% in the fit window and is tested on ω_r only), justified by the pilot error ($\leq 0.05\%$ and $\leq 2\%$) plus a fit-window systematic of about 1%.

Landau damping is a pure energy transfer from the coherent field to resonant particles, so any spurious numerical dissipation or heating would masquerade as, or mask, it; energy bookkeeping is therefore the primary diagnostic. *Energy criteria:* (E1) $|\Delta\mathcal{U}/\mathcal{U}| \leq 10^{-6}$ in every run; (E2) a cold control ($\sigma_p = 0$, no resonant particles) shows $|\gamma| \leq 10^{-4}m$; (E3) the perturbation energies $W_A = \frac{1}{2}(\mathcal{U}_A^+ + \mathcal{U}_A^-) - \mathcal{U}_A^0$ and W_ϕ (same for matter) satisfy $|\Delta(W_A + W_\phi)| \leq 10^{-2} \max W_A$, i.e. the field energy lost is found in the matter.

2.2.3 Cyclotron resonance at $B \sim B_c$ (primary diagnostic: dispersion fidelity)

A uniform field $B_z = B$ enters the transverse links through the non-dynamical potential $A_2^{\rm ext} = -Bx$, with flux quantisation $qBL\Delta_\perp = 2\pi$. A uniform magnetised plasma requires many guiding centres; we therefore let the matter consist of N_y transverse-momentum species $p_f = 2\pi f/(N_y\Delta_\perp)$, which is the exact restriction of the two-dimensional lattice action to y-independent gauge fields (cross-species terms average to zero over the transverse cell). Their Landau orbits are centred at $x_f = fL/N_y$, so the density is uniform. Each species starts in the lowest lattice Landau level (computed by sparse diagonalisation of the lattice operator), and a uniform kick $e^{i(2\pi/L)x}$ launches coherent cyclotron motion; the frequency of the uniform current $\bar{J}_1(t)$ is the cyclotron line ω_c, and the phase velocity of the ground-state projection gives the rest energy E_0 of the lowest level. For a spin-0 field the continuum Landau levels are $E_n^2 = m^2 + (2n+1)|qB|$, so

$$\omega_c^{(0)} = \sqrt{m^2 + 3|qB|} - \sqrt{m^2 + |qB|}, \qquad E_0^{(0)} = \sqrt{m^2 + |qB|}, \tag{9}$$

whereas for spin-$\frac{1}{2}$ electrons with $g = 2$ the levels are $E^2 = m^2 + 2n|qB|$ for the spin projection that lowers the energy [7], so that

$$\omega_c^{(1/2)} = \sqrt{m^2 + 2|qB|} - m, \qquad E_0^{(1/2)} = m. \tag{10}$$

At $B = B_c$, $\omega_c^{(0)} = 0.586m$ and $\omega_c^{(1/2)} = 0.732m$; for $qB \ll m^2$ the relative difference is $qB/2m^2$, so it exceeds 1% already at $B \approx 0.02B_c \approx 10^8$ T. The plasma is dilute ($\omega_p = 0.01m$), so that collective shifts, of relative size $(\omega_p/\omega_c)^2 \approx 3 \times 10^{-4}$, are negligible. *Pre-stated failure criterion:* the unmodified scheme fails if $|\omega_c^{\rm sim}/\omega_c^{(1/2)} - 1| > 1\%$ and $|E_0^{\rm sim}/m - 1| > 1\%$ at *every* resolution; it is simultaneously required to agree with Eq. (9) to within 1%, which certifies that the discrepancy is not an implementation error. *Repair criterion:* the Pauli-coupled scheme of Sec. 2.3.1 must satisfy $|\omega_c^{\rm sim}/\omega_c^{(1/2)} - 1| \le 1\%$ and $|E_0^{\rm sim}/m - 1| \le 1\%$ at every resolution. The 1% tolerance is twice the sum of the estimated transverse-cell anharmonicity ($\Delta_\perp/\ell_B = 0.157$, $\approx 0.2\%$), the x-discretisation error ($\le 0.2\%$ at the coarsest grid), the collective shift and the spectral-fit error.

2.2.4 Spontaneous pair creation by a Sauter pulse (primary diagnostic: non-trivial ϕ–A coupling)

A homogeneous pulse $E(t) = E_0 \,\mathrm{sech}^2[(t - t_c)/\tau]$ is imposed through a uniform external link potential $A_1^{\rm ext}(t) = E_0\tau \tanh[(t - t_c)/\tau]$, which enters only the Wilson lines; the matter fields are evolved as test fields. A mode of canonical momentum p has kinetic momentum $P(t) = p - qA_1(t)$ and obeys $\ddot{u} + [m^2 + P(t)^2]u = 0$; pair creation is entirely due to the non-adiabatic, non-perturbative dependence of the Wilson phase $e^{-iq\Delta x A_1}$ on time. The connection problem is again hypergeometric and gives the Bogoliubov coefficient [82, 12]

$$|\beta_p|^2 = \frac{\cosh\left[\pi\sqrt{4(qE_0\tau^2)^2 - 1}\right] + \cosh[\pi\tau(\omega_+ - \omega_-)]}{2\sinh(\pi\tau\omega_+)\sinh(\pi\tau\omega_-)}, \tag{11}$$

with $\omega_\pm = [m^2 + (p \pm qE_0\tau)^2]^{1/2}$, which tends to the Schwinger factor $e^{-\pi m^2/qE_0}$ for $\tau \to \infty$; we verified it against direct integration to 10^{-10}. We use $qE_0 = m^2$ ($E_0 = E_c$), $\tau = 4/m$, $L = 50/m$, and all lattice modes with $|p| \le 8m$. The out-particle number n_p is obtained by projecting the fields on exact lattice positive- and negative-frequency out-modes. At tree level the vacuum is $\phi = 0$, a fixed point of Eq. (3); if classical noise of occupation ϵ per mode is added, linearity gives an out-occupation $\epsilon(1 + 2|\beta_p|^2)$, i.e. a yield $2\epsilon|\beta_p|^2$ proportional to the seed. *Pre-stated failure criterion:* the original prescription (classical vacuum plus the $\sigma(A) = 10^{-4}m$ link noise of Ref. [26]) fails if the spontaneous pair density is $\le 10^{-6}$ of the prediction at every resolution, and classically seeded ensembles must show a yield proportional to ϵ. *Repair criterion* (Sec. 2.3.2): $|n^{\rm sim}/n - 1| \le 1\%$ for the pair density $n = \int dp\,|\beta_p|^2/2\pi$ and $\max_p |n_p^{\rm sim} - |\beta_p|^2| \le 10^{-2}\max_p |\beta_p|^2$ at every resolution, with each flavour's charge conserved to 10^{-12}.

2.3 Two gauge-invariant extensions of the discrete action

2.3.1 Lattice Pauli coupling

To give the matter a spin magnetic moment without leaving the variational framework we promote each species to a two-component field $\psi = (\psi_\uparrow, \psi_\downarrow)$ and add to Eq. (2)

$$S_P = \Delta t\, \Delta x\, \Delta_\perp^2 \sum_{n,i} \sum_f w_f\, \mu\, \psi_{f,i}^{n\dagger} (\sigma_y \bar{B}_{y,i}^n + \sigma_z \bar{B}_{z,i}^n) \psi_{f,i}^n, \tag{12}$$

where $\bar{B}_{z,i}^n = \frac{1}{2}(B_{z,i+1/2}^n + B_{z,i-1/2}^n) + B_z^{\text{ext}}$ (and similarly $\bar{B}_y$), $\mu = gq/2$ and $g = 2$, i.e. the magnetic part of the second-order (Feynman–Gell-Mann) form of the Dirac operator, $(D_\mu D^\mu + m^2) - \frac{q}{2}\sigma^{\mu\nu}F_{\mu\nu}$ [83]. The face-centred magnetic field is averaged to vertices, so S_P is a sum of products of gauge-invariant quantities. Variation with respect to $\bar{\psi}$ adds $\Delta t^2 \mu(\sigma \cdot \bar{\mathbf{B}})\psi^n$ inside the bracket of Eq. (3); variation with respect to A_2 and A_3 adds the lattice magnetisation currents $\mu(M_{z,i+1} - M_{z,i-1})/2\Delta x$ and $\mu(M_{y,i-1} - M_{y,i+1})/2\Delta x$ to J_2 and J_3 in Eq. (4), with $\mathbf{M} = \sum_f w_f \psi_f^\dagger \boldsymbol{\sigma} \psi_f$. In a uniform field the spin-up component then obeys Eq. (3) with $m^2 \to m^2 - \mu B$, so the continuum spectrum becomes $E^2 = m^2 + (2n+1)|qB| - qB\sigma_z$, the Dirac spectrum.

Invariants. (1) S_P is invariant under the lattice $U(1)$ gauge transformation, and it contains no time-like or x-links, so the Noether charge and current are those of Eq. (5): local charge conservation and the Gauss law hold exactly, as before. (2) The Bianchi identities are untouched ($\mathbf{B}$ is still computed from the links). (3) The magnetisation currents are discrete curls and drop out of the continuity equation. (4) The discrete energy acquires the time-centred magnetisation term $-\mu\,\mathrm{Re}\, e^{iq\Delta t A_0}\psi^{n\dagger}(\sigma \cdot \bar{\mathbf{B}}^{n+1/2})\psi^{n+1}$ and is conserved to the same $\mathcal{O}(\Delta^2)$ accuracy. *Relaxed:* Lorentz covariance of the spin coupling—the electric part of $\sigma^{\mu\nu}F_{\mu\nu}$, responsible for spin–orbit and electric-dipole effects, is omitted, so the extension is exact for static magnetic fields and approximate when $E \sim B$ in the particle frame; spin–statistics remains bosonic.

2.3.2 Half-quantum Bogoliubov extension

The quantum field $\hat{\phi} = \sum_k (\hat{a}_k u_k + \hat{b}_{-k}^\dagger v_k)$ evolves linearly in a given gauge field, so any Gaussian state is fully described by the classical mode functions u_k, v_k, which obey the same discrete Klein–Gordon equation. We therefore replace the single classical field by $2N_k$ flavours $\phi_{k,\pm}$ and use the action

$$S_{\text{hq}} = \sum_{k,s=\pm} w\, S_{\text{KG}}[\phi_{k,s}; A] + S_{\text{EM}}[A], \qquad w = \frac{1}{2L\Delta_\perp^2}, \tag{13}$$

where S_{KG} and S_{EM} are the matter and field parts of Eq. (2) and the flavours are initialised as exact lattice plane waves of unit charge density, $\phi_{k,\pm}^0 = \nu_k^{1/2} e^{ikx}$, $\phi_{k,\pm}^1 = \phi_{k,\pm}^0 e^{\mp i\omega_k \Delta t}$, $\nu_k = \Delta t/2\sin(\omega_k \Delta t)$, with ω_k the lattice frequency. The weight w assigns one half quantum per mode to the volume $L\Delta_\perp^2$, so the Maxwell source is the symmetrised expectation value of the current in the instantaneous state (a one-loop, Hartree-type back-reaction [10, 11]). Each flavour obeys Eq. (3) unchanged; the Maxwell equations are sourced by the weighted sums (5). Observables are symmetrised expectation values, e.g. $n_k = \sum_s wL\Delta_\perp^2 |\langle u_k^{\text{out}} | \phi_{k,s}\rangle|^2 - \frac{1}{2}$. For an external field this construction is *identical* in expectation to sampling a single classical field from the Wigner distribution of the vacuum, $\phi = \sum_k (a_k u_k + b_k^* v_k)$ with $\langle |a_k|^2 \rangle = \langle |b_k|^2 \rangle = \frac{1}{2}$ [24]; we use both forms.

Invariants. (1) Under a local gauge transformation all flavours rotate together, so S_{hq} is gauge invariant and the weighted current is exactly conserved; the Gauss law is preserved. (2) S_{hq} is in addition invariant under independent global phase rotations of every flavour, so each flavour's Klein–Gordon charge is conserved separately; since the flavour equations are linear, the discrete Klein–Gordon inner product between any two flavours is conserved, which enforces the Bogoliubov normalisation $|\alpha_k|^2 - |\beta_k|^2 = 1$ exactly. (3) The Bianchi identities and the form of the energy law (6) are unchanged; the half-quantum vacuum energy is a constant offset. *Relaxed:* the gauge field is coupled to the expectation value of the current only (mean-field); matter–matter correlations beyond Gaussian order and gauge-field fluctuations are dropped. The cost grows from N to $2N_k N$ per step.

3 Experimental setup

All simulations use the implementation of Sec. 2.1 in double precision (`proposal/code/`; Python 3.12, NumPy 2.2, Numba 0.68, 32 threads). Every run writes its configuration, random seeds, UTC timestamps, conservation time series and observables to `proposal/logs/`; the run scripts are listed in `code/runs/production_*.sh`, and every number quoted below is generated from these logs by `code/analysis/make_macros.py`. Resolutions are always $\Delta t/\Delta x \le 1/2$ except where the original paper prescribes otherwise.

Reproduction of the linear-wave spectra [26]. Cold single-species plasma with immobile ions, $\omega_p = 0.85m$, $q = 0.3$, white link noise $\sigma(A) = 10^{-4}m$, $\phi^0 = \sqrt{n_0/2m}e^{i\alpha}$, $\phi^1 = \phi^0 e^{-im\Delta t}$ with random α, temporal gauge, 100 members. (a) The paper's parameters: 512 points, $m\Delta x = 0.04$, $m\Delta t = 0.02$, 1024 time levels. (b) For a quantitative test, the same physical box ($L = 20.48/m$) with a record of $163.84/m$ (Hann window, frequency bin $0.038m$) at $m\Delta x = 0.08, 0.04, 0.02$ ($m\Delta t = \Delta x/2$). Peak frequencies of the ensemble-averaged E_y (transverse) and E_x (Langmuir branch, $0.3m < \omega < 1.9m$) power spectra are refined by a parabolic fit and compared with Eqs. (21)–(23) of Ref. [26] for $0 < k\Delta x \le 0.5$. *Tolerance:* maximum relative deviation $\le 2\%$ (lattice dispersion $\le 1\%$ at $k\Delta x = 0.5$ plus 1% peak-location uncertainty).

Reproduction of the laser–plasma demonstration [26]. Plasma slab with plateau density $n_0 = m^3$ ($\omega_p = 0.3m$) on $350 \le mx \le 450$ and Gaussian ramps of width $20/m$, immobile ions, box $L = 720/m$, periodic boundaries, temporal gauge. Following the paper, the slab wave function $\sqrt{n_0(x)/2m}\, e^{-imt}$ is first evolved alone for $T_{\rm mix} = 100/m$ and then sampled at 200 random times in $[100, 200]/m$; each sample receives a random global phase and is combined with a linearly polarised pulse $A_y \propto e^{-\xi^2/2\tau^2}\cos\omega_0\xi$, $\xi = x - x_0 - t$, $\omega_0 = 0.7m$, $\tau = 20/m$, $x_0 = 160/m$, normalised amplitude $a_0 = qA/m = 1$ or 16, with the second time level computed from the exact lattice vacuum dispersion. White noise $10^{-4}m$ is added to the transverse links. Resolution $m\Delta x = 0.04$, $m\Delta t = 0.005$ (1.8×10^4 points, 10^5 steps) as in the paper. Additional runs with 32 members: an intensity scan $a_0 = 2, 4, 8$, and resolution studies at $m\Delta x = 0.08$ ($m\Delta t = 0.01$) and $m\Delta x = 0.02$ ($m\Delta t = 0.0025$) for $a_0 = 1$ and 16. *Criteria* (from the values reported in Ref. [26]): charge conserved to $\le 10^{-12}$; energy conserved to $\le 1.2 \times 10^{-5}$ ($a_0 = 1$) and $\le 4 \times 10^{-3}$ ($a_0 = 16$), i.e. twice the published 6 ppm and 0.2%; for $a_0 = 1$ the wakefield stays below E_c and the excess positive charge ("positrons") stays below 10^{-3} of the ion charge, while for $a_0 = 16$ the field exceeds E_c and the positron fraction exceeds 10^{-2}.

Superradiant Klein reflection. Box $L = 600/m$, step at $x_a = 300/m$ (the potential returns to zero at $560/m$, outside the measurement window), packet centred at $240/m$ with spatial width $10/m$ and peak amplitude 10^{-3}, final time $300/m$, reflected charge integrated over $x < 275/m$. Resolution ladder $m\Delta x = 0.1, 0.05, 0.025, 0.0125$ with $\Delta t = \Delta x/2$, five energies per run, three gauges per resolution. An 8-member ensemble at $m\Delta x = 0.05$, $\varepsilon = 1.5m$ uses random global phases, random sub-cell packet positions and $10^{-4}m$ transverse link noise.

Landau damping. Box $L = 192\pi/m \approx 603/m$ (so that the five wavenumbers $k = 2\pi j/L$, $j = 30, 36, 42, 48, 54$, are exact lattice modes), 289 species with $|p_j| \le 6\sigma_p$, final time $400/m$. Resolution ladder $N = 1536, 3072, 6144, 12288$ ($m\Delta x = 0.39$–0.049, $\Delta t = \Delta x/2$); a doubled box ($L \approx 1206/m$, 577 species) at $m\Delta x = 0.196$ tests the effect of the discrete species spacing; a cold control ($\sigma_p = 0$) and a single-wavenumber run for the energy budget use $N = 3072$. For comparison we also evolve the same plasma as a single classical field with random phases (64 realisations, $\pm\epsilon$ and 0), the representation of a thermal state in the classical-statistical approximation.

Magnetised plasma. $qB = m^2$ ($\ell_B = 1/m$), $L = 40/m$, $N_y = 64$ transverse-momentum species ($\Delta_\perp = 0.157/m$, guiding-centre spacing $0.625/m$), $\omega_p = 0.01m$, final time $400/m$, temporal gauge. Resolution ladder $m\Delta x = 0.125, 0.0625, 0.03125, 0.015625$; transverse refinement $L = 80/m$ and $160/m$ ($N_y = 128, 256$, $\Delta_\perp/\ell_B = 0.079, 0.039$); four random-phase realisations; and a field scan $qB/m^2 = 0.1, 0.3, 3$ at fixed $\Delta x/\ell_B = 1/16$ and $\Delta_\perp/\ell_B = 0.157$. Each configuration is run with the original scheme ($S = 1$) and with the Pauli extension ($S = 2$, $g = 2$, spin-polarised along $+z$).

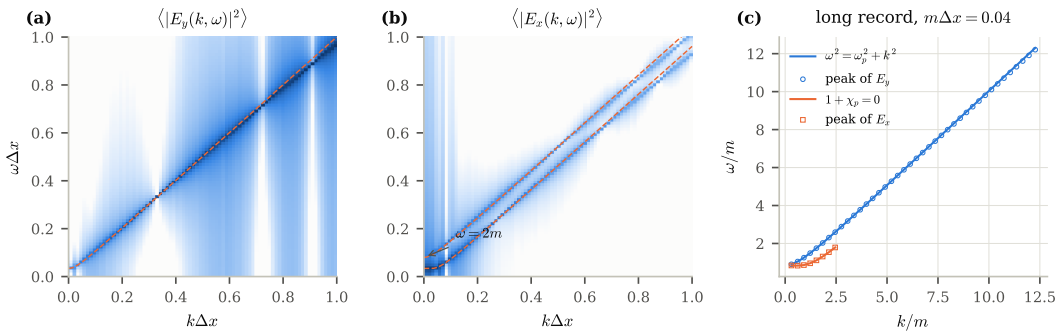


Figure 1: Reproduction of the linear-wave spectra of Ref. [26] (their Fig. 4). (a,b) Ensemble-averaged power of E_y and E_x with the paper's parameters ($m\Delta x = 0.04$, $m\Delta t = 0.02$, 1024 levels, 100 members; logarithmic colour scale over six decades), with the tree-level dispersion relations (dashed). (c) Peak frequencies from the longer, Hann-windowed records at $m\Delta x = 0.04$ against Eqs. (21)–(23) of Ref. [26].

Vacuum pair creation. Box $L = 50/m$, $qE_0 = m^2$, $\tau = 4/m$, pulse centre $t_c = 48/m$, final time $96/m$, modes $|p| \leq 8m$ (127 modes), resolution ladder $m\Delta x = 0.1, 0.05, 0.025, 0.0125$. Each resolution is run with the original prescription and with the half-quantum extension; additional extension runs use $qE_0 = 0.5m^2$ and $2m^2$. Classical-noise ensembles with seed occupations $\epsilon = 0.01$ and 0.1 (400 members) and the Wigner ensemble $\epsilon = \frac{1}{2}$ (1000 members) use $m\Delta x = 0.05$ and the control-variate estimator $n_p = \langle |b_p^{\mathrm{out}}|^2 - |b_p^{\mathrm{in}}|^2 \rangle$. For the back-reaction runs the uniform field is dynamical: initial $qE_0 = 2m^2$, no external drive, all $2N$ lattice modes as flavours with $w = 1/(2L\Delta_\perp^2)$ and transverse cell area $\Delta_\perp^2 = 0.1/m^2$, $L = 25.6/m$, final time $40/m$, $m\Delta x = 0.1, 0.05, 0.025$.

4 Experimental results

Table 1 summarises all six settings; the subsections below give the details. Conservation columns quote the largest relative deviation over the run for the realisation shown in Fig. 7 (absolute values for states with zero total charge).

4.1 Reproduction of the linear-wave spectra

With the parameters of Ref. [26] (512 points, $m\Delta x = 0.04$, 1024 time levels, 100 members) the ensemble-averaged spectra reproduce their Fig. 4: the transverse power follows $\omega^2 = \omega_p^2 + k^2$ up to $k\Delta x \sim 1$, and the longitudinal power is concentrated on the zero-frequency band, the Langmuir branch and the pair branch starting at $\omega = 2m$ (Fig. 1a,b). Total charge is constant to 1.6×10^{-14} and the discrete energy to 8.4×10^{-14} in every member. The quantitative test uses the longer records (Fig. 1c): over all 40 resolved wavenumbers with $0 < k\Delta x \leq 0.5$ the transverse peak frequencies deviate from Eq. (21) of Ref. [26] by at most 0.72%, 0.75% and 0.77% at $m\Delta x = 0.08, 0.04, 0.02$, and the Langmuir peaks deviate from Eqs. (22)–(23) by at most 0.15%, 0.07% and 0.07%, all within the pre-stated 2%. At fixed k the deviation decreases with Δx (Fig. 8a); the maximal values are reached at $k\Delta x = 0.5$, where they are dominated by the known lattice dispersion of the Yee-type field term.

4.2 Reproduction of the wakefield-to-pair-production transition

Figure 2 shows the ensemble-averaged charge and electromagnetic energy densities for $a_0 = 1$ and $a_0 = 16$ (200 members each), to be compared with Figs. 5–7 of Ref. [26]. For $a_0 = 1$ the pulse excites a wake whose largest longitudinal field is $\max E_x = 0.21\, E_c$ (member-to-member standard deviation $1.2 \times 10^{-4}\, E_c$), below the wave-breaking estimate $a_0 m\omega_p/q = 0.3E_c$ quoted in Ref. [26]; electrons are expelled backwards from the left boundary and forward beams form behind the pulse; the excess positive charge never exceeds 2.2×10^{-4} of the ion charge (no pair creation); the pulse centroid follows the geometric-optics trajectory $\dot{x} = (1 - \omega_p^2(x)/\omega_0^2)^{1/2}$ to

Table 1: Main results. Predictions, tolerances and failure criteria were fixed in `logs/tolerances.json` before the production runs (Sec. 2.2). "Reproduction" rows repeat the two demonstrations of Ref. [26] in our implementation; "validated regime" rows test the scheme where it should be exact; "fundamental limit" rows test premises of the scheme and the extensions of Sec. 2.3. $|\Delta Q/Q|$ and $|\Delta\mathcal{U}/\mathcal{U}|$ are maxima over the run; for the vacuum, $Q = 0$ and $|Q|$ is quoted, and the energy is not conserved under an external pulse (n/a); the back-reaction run conserves energy to the value given in the outcome.

Test	Category	Regime probed	Prediction	Tolerance / failure criterion	Outcome	$\lvert\Delta Q/Q\rvert$	$\lvert\Delta\mathcal{U}/\mathcal{U}\rvert$
Linear waves (Ref. [26], Sec. III.1)	reproduction	cold, unmagnetised, single species; 1D dispersion	$\omega^2 = \omega_p^2 + k^2$; $1 + \chi_p = 0$	peak frequency within 2% for $k\Delta x \leq 0.5$	max. deviation 0.75% (transverse), 0.07% (Langmuir) at $m\Delta x = 0.04$; passes at all three resolutions	5.0×10^{-15}	7.4×10^{-14}
Laser–plasma slab (Ref. [26], Sec. III.2)	reproduction	1D laser–plasma at $a_0 = 1, 16$	wakefield $< E_c$ without pairs at $a_0 = 1$; pair creation above the Schwinger threshold at $a_0 = 16$	$\max E_x \lesssim E_c$; positron fraction $< 10^{-3}$ / $> 10^{-2}$; $\Delta\mathcal{U} \leq 2\times$ published	$\max E_x = 0.21\,E_c$, positrons 2.2×10^{-4} ($a_0 = 1$); $1.97\,E_c$, 7.7×10^{-2}, matter energy $\times 14.0$ ($a_0 = 16$); passes at all three resolutions, but $a_0 = 16$ yields change by 7%–18% from $m\Delta x = 0.04$ to 0.02	9.0×10^{-14}, 7.5×10^{-14}	1.2×10^{-6}, 3.5×10^{-4}
Klein step	validated regime	static supercritical step ($qE = 2m^2$), stimulated pair creation	R of Eq. (7) ($R > 1$)	$\lvert R_{\rm sim}/R - 1\rvert \leq 10^{-3}$; gauge difference $\leq 10^{-10}$	max. deviation 9.8×10^{-6}; gauges agree to 7.6×10^{-12} with link phases winding up to 5.3 turns; passes at all resolutions	3.7×10^{-14}	6.2×10^{-11}
Warm plasma, Landau damping	validated regime	$k\lambda_D = 0.4$–0.7, quantum recoil, $\gamma_{\rm res} = 1.16$	roots of $1 + \chi = 0$, Eq. (8)	$\lvert\delta\omega_r\rvert \leq 0.5\%$, $\lvert\delta\gamma\rvert \leq 5\%$; E1–E3	max. $\lvert\delta\gamma\rvert = 0.24\%$, $\lvert\delta\omega_r\rvert = 6.8 \times 10^{-5}$ ($N = 12288$); passes for $N \geq 3072$, coarsest grid outside (16.04%); cold control $\lvert\gamma\rvert \leq 3.7 \times 10^{-6}$; budget closes to 4.9×10^{-3}	3.5×10^{-16}	1.3×10^{-10}
Magnetised plasma at $B = B_c$	fundamental limit	Landau quantisation, $\hbar\omega_c \sim m$	spin $\frac{1}{2}$: $\omega_c = 0.732m$, $E_0 = m$	fails if $> 1\%$ off at every Δx; repair within 1%	spin 0: 20.2% (ω_c), 41.4% (E_0) off at all Δx (matches spin-0 theory to 0.22%); Pauli term: 0.19%, 0.08%	3.2×10^{-14}	3.2×10^{-10}
Vacuum under a Sauter pulse	fundamental limit	homogeneous pulse, $qE_0 = m^2$, empty initial state	$n = \int dp\,\lvert\beta_p\rvert^2/2\pi$, Eq. (11)	fails if $n_{\rm sim} \leq 10^{-6} n$ at every Δx; repair within 1%	classical vacuum: $n_{\rm sim} = 0$ at all Δx; half-quantum: 7.6×10^{-7}; back-reaction energy error 1.2×10^{-3}	1.2×10^{-12} (abs.)	n/a

within $1.6/m$; and the transmitted spectrum carries Raman sidebands at $\omega_0 + n\omega_p$ up to $n = 8$, as reported in Ref. [26]. The laser loses 6.2% of its energy to the plasma. For $a_0 = 16$ the longitudinal field reaches $1.97\,E_c$, pair creation sets in once the pulse has penetrated the density plateau (the excess positive charge exceeds 10^{-3} of the ion charge at $mt \approx 251$, as in Fig. 5b of Ref. [26]), the positive charge reaches 7.7×10^{-2} of the ion charge and ends at 5.1×10^{-2}, and the matter energy grows by a factor 14.0 (from 150.1 to $2102.9\,m^3$ per transverse area) while the total energy stays constant; positron-rich outflows leave the slab from its left boundary (Fig. 2a, right). Charge is conserved to 2.1×10^{-13} and 1.2×10^{-13}, and energy to 1.2×10^{-6} ($a_0 = 1$) and 3.5×10^{-4} ($a_0 = 16$), better than the 6 ppm and 0.2% reported in Ref. [26]. All four pre-stated criteria are met. The intensity scan (Fig. 2c) locates the transition: $\max E_x/E_c = 0.21, 0.42, 0.77, 1.47, 1.97$ and $\max Q_+/Q_{\rm ion} = 2.2\times10^{-4}, 5.1\times10^{-3}, 2.4\times10^{-2}, 6.5\times10^{-2}, 7.7\times10^{-2}$ for $a_0 = 1, 2, 4, 8, 16$. The positron fraction rises by more than an order of magnitude between $a_0 = 1$ and $a_0 = 2$, i.e. when the amplitude $a_0\omega_0 m/q$ of the laser field itself crosses E_c ($a_0 = m/\omega_0 \approx 1.43$, the threshold quoted in Ref. [26]), whereas the wakefield crosses E_c only between $a_0 = 4$ and 8; pair creation in the slab is therefore controlled by the laser field acting on the dense plasma rather than by the longitudinal wakefield alone. The 32-member resolution study (Fig. 8b) shows that the $a_0 = 1$ run

is converged at the paper's resolution: at $m\Delta x = 0.08, 0.04, 0.02$ the largest wakefield is 0.211, 0.21 and $0.211\,E_c$, the final matter energy 156.1, 156.1 and $156.1\,m^3$, and the energy error falls as Δx^2 (4.8×10^{-6}, 1.2×10^{-6}, 3.0×10^{-7}). For $a_0 = 16$ every resolution shows copious pair creation and meets the pre-stated criteria (energy errors 4.5×10^{-4}, 3.5×10^{-4}, 1.5×10^{-4}; charge to 3.1×10^{-13}), but the yields are not converged at the paper's resolution: the largest positron fraction is 4.7×10^{-2}, 7.7×10^{-2} and 9.0×10^{-2}, the final matter energy 1359.5, 2102.9 and $2246.7\,m^3$, and the largest wakefield 1.876, 1.97 and $2.134\,E_c$. With the pulse normalisation used here (see below), the $a_0 = 16$ yields are therefore not converged at the resolution that Ref. [26] describes as converged; for a lower-energy pulse the residual may be smaller. The residual is longitudinal: shrinking only the transverse cell at $m\Delta x = 0.04$, to $\Delta_\perp = 0.02/m$ and $0.01/m$, leaves the positron fraction at 7.7×10^{-2} and 7.6×10^{-2} and the matter energy at 2056.0 and $2092.0\,m^3$. It is consistent with the kinetic momenta reached by the matter field: the charge-weighted mean of the local kinetic lattice momentum, estimated from $|D_1\phi|^2\Delta x^2/4|\phi|^2 = \sin^2(P\Delta x/2)$, peaks at 5.4, 6.7 and $7.5\,m$ on the three grids (it grows with resolution), and the fraction of the matter with $|P|\Delta x > \pi/2$ peaks at 4.47%, 0.49% and 0.07%, whereas for $a_0 = 1$ the mean kinetic momentum stays at $0.18\,m$. Strongly driven runs therefore need $|P|\Delta x \ll 1$ for the accelerated particles, the condition discussed in Sec. 5.

One quantitative difference with Ref. [26] concerns the pulse energy. With the pulse formula and parameters stated there ($\tau = 20/m$, $a_0 = qA/m$), the initial laser energy per transverse area is $96.7\,m^3$ for $a_0 = 1$ and $24765.4\,m^3$ for $a_0 = 16$, three to four times larger than the values that can be read from their Fig. 7b; the same paper also states an intensity full width of "about twice the plasma skin depth" ($\approx 7/m$), which is inconsistent with $\tau = 20/m$ but would account for the factor. We kept the explicit formula. The qualitative sequence—wakefield without pairs at $a_0 = 1$, pair creation with strong depletion at $a_0 = 16$—and all conservation properties are reproduced; the absolute fraction of laser energy converted (7.9% here) depends on this normalisation.

4.3 Superradiant Klein reflection from a supercritical step

The packets reach the step at $mt \approx 60$–110, depending on their group velocity; afterwards a reflected packet carrying *more* charge than the incident one moves back, while a packet of opposite charge—antiparticles with negative kinetic energy—propagates into the step (Fig. 3b). For $\varepsilon = 1.5m$ the transmitted charge is -0.2327 times the incident charge, so that the total charge is conserved while the reflected charge exceeds the incident one by 23.3%; the total charge is constant to 9.5×10^{-14} throughout. At the finest resolution the measured reflection coefficients are $R = 1.15190, 1.23268, 1.27645, 1.23212$ for $\varepsilon/m = 1.2, 1.5, 2.0, 2.5$, against the packet-averaged predictions of Eq. (7), $1.15191, 1.23269, 1.27645, 1.23211$, i.e. relative deviations of 9.8×10^{-6}, 5.8×10^{-6}, 2.1×10^{-6} and 9.1×10^{-6} (Fig. 3a). The largest Klein-zone deviation is 3.9×10^{-4}, 1.0×10^{-4}, 2.7×10^{-5} and 9.8×10^{-6} at $m\Delta x = 0.1, 0.05, 0.025, 0.0125$, well inside the pre-stated 10^{-3} at every resolution and decreasing approximately as Δx^2 (Fig. 8c). The eight-member ensemble at $m\Delta x = 0.05$ gives $R = 1.232660 \pm 1.2\times10^{-9}$ (prediction 1.232687): random global phases, sub-cell packet offsets and $10^{-4}m$ link noise leave R unchanged at the level of the pre-stated tolerance. The exponentially small normal-zone reflection ($\varepsilon = 5.5m$, $R \approx 10^{-3}$) converges more slowly, with relative deviations 49%, 10%, 3% and 1% along the ladder; it is controlled by the lattice dispersion at $k\Delta x \approx 0.5$ and is sensitive to the smoothness of the step on the grid scale.

Gauge invariance, the primary diagnostic, holds to round-off. In the temporal gauge the link variable inside the step grows linearly in time and the Wilson phase $q\Delta x A_1$ winds up to 9.5, 4.8, 2.4 and 1.2 times around the circle at the four resolutions; after the random gauge transformation it winds up to 12.6–5.3 times. Nevertheless the charge density, electric field, $|\phi|^2$ and R agree with the static-gauge evolution to a maximal relative difference of 7.1×10^{-13}, 1.0×10^{-12}, 3.5×10^{-12} and 7.6×10^{-12}, far inside the pre-stated 10^{-10}. A supercritical field that is sustained in time but bounded in space is therefore represented exactly by the compact links, however large the vector potential becomes: the unbounded growth of A in the temporal gauge, flagged as a possible concern in Ref. [26], is harmless. Any error in a Wilson-line phase, in the sign of the seagull term or in the Lorenz update would produce gauge-dependent results at order unity in this test.

4.4 Landau damping with quantum recoil in a warm relativistic plasma

The odd response $\delta E_k(t)$ decays exponentially after a short transient—for the most strongly damped mode by 12 e-folds within the fit window, before it reaches the floor set by higher-order terms—at

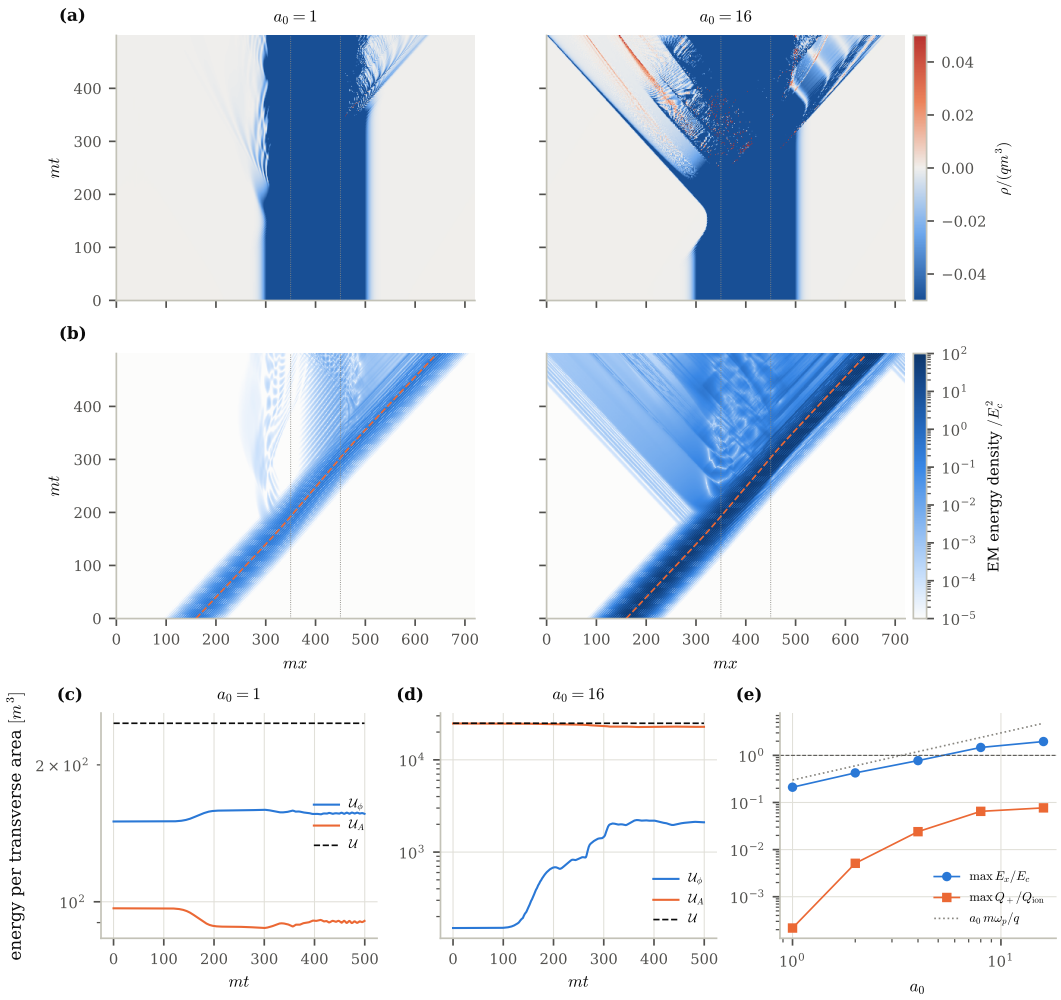


Figure 2: Reproduction of the wakefield-to-pair-production transition of Ref. [26] (their Figs. 5–7). (a) Ensemble-averaged charge density (200 members; negative: excess "electrons", positive: excess "positrons"), clipped at $\pm 0.05\, qm^3$ as in the original, for $a_0 = 1$ (left) and 16 (right). (b) Electromagnetic energy density in units of E_c^2, with the geometric-optics trajectory of the pulse centre (dashed); dotted lines mark the density plateau. (c,d) Matter and field energies per transverse area for $a_0 = 1$ and 16. (e) Intensity scan of the largest wakefield and of the largest excess positive charge ("positron" fraction).

rates that differ by a factor of 70 across the five wavenumbers (Fig. 4a). At the finest resolution ($N = 12288$) the fitted complex frequencies are listed in Table 2: the real frequencies agree with the roots of Eq. (8) to within 6.8×10^{-5} and the damping rates of the four modes with $\gamma/\omega_r \geq 0.02$ to 0.24%, 0.14%, 0.11% and 0.13%, inside the pre-stated 0.5% and 5%; even the least-damped mode, which is not subject to the γ criterion, is reproduced to 4.7×10^{-4}. The largest damping-rate deviation is 16.04%, 4.01%, 1.00% and 0.24% for $N = 1536$, 3072, 6144 and 12288, and the largest frequency deviation 0.20%, 5.0×10^{-4}, 1.3×10^{-4} and 6.8×10^{-5} (Fig. 8d). On the coarsest grid ($m\Delta x = 0.39$, where the fastest species have $p\Delta x \approx 0.6$) the damping rates are therefore outside the 5% tolerance; from $N = 3072$ on they are inside it, and the deviation falls by a factor of about four per halving of Δx, as expected from the second-order lattice dispersion of the resonant particles. Doubling the box (twice as many species, half the species spacing) changes the fitted rates by at most 2.3×10^{-7} (relative) and the frequencies by 1.1×10^{-9}: for a Gaussian distribution the species sum converges exponentially to the momentum integral within the recurrence time, so the residual at finite Δx is purely a lattice-dispersion effect. Four realisations with different random kick and species phases ($N = 3072$) give $\gamma = 0.02858 \pm 1.0 \times 10^{-6}$ at $k = 0.5m$ (mean $\pm$ standard deviation),

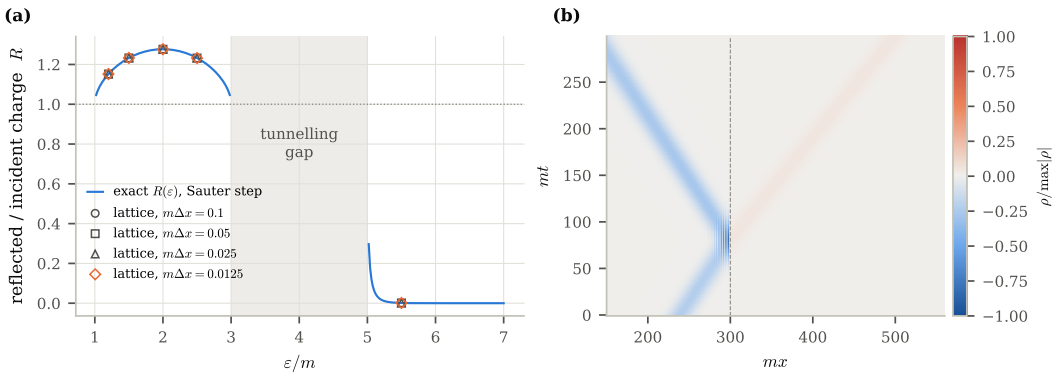


Figure 3: Superradiant Klein reflection from a Sauter step with $V_0 = 4m$, $\ell = 1/m$ ($qE_{\max} = 2m^2$). (a) Reflected-to-incident charge ratio: exact result, Eq. (7) (line; shaded: tunnelling gap $|\varepsilon - V_0| < m$), and lattice results at four resolutions (symbols; the finest is highlighted). (b) Space–time charge density for $\varepsilon = 1.5m$ ($m\Delta x = 0.025$, static gauge): the reflected packet carries more charge than the incident one, and a packet of opposite charge enters the step (dashed line: step centre).

Table 2: Langmuir-wave complex frequencies (units of m) in the warm plasma ($\omega_p = 0.2m$, $\sigma_p = 0.25m$): tree-level scalar-QED prediction [roots of Eq. (8)], lattice simulation at $N = 12288$, and the damping rates predicted by three reduced kinetic models with the same momentum distribution (non-relativistic quantum, relativistic classical and classical Vlasov). The γ tolerance applies to $k \geq 0.375m$.

k/m	ω_r (theory)	ω_r (lattice)	γ (theory)	γ (lattice)	$\gamma_{\text{NR quantum}}$	$\gamma_{\text{rel. classical}}$	γ_{Vlasov}
0.3125	0.23447	0.23446	0.00065	0.00065	0.0152	0.0005	0.0119
0.3750	0.25845	0.25844	0.00581	0.00580	0.0320	0.0046	0.0246
0.4375	0.28592	0.28592	0.01571	0.01569	0.0541	0.0129	0.0406
0.5000	0.31630	0.31629	0.02910	0.02907	0.0811	0.0242	0.0589
0.5625	0.34966	0.34963	0.04502	0.04496	0.1124	0.0381	0.0793

showing that the fitted rates are insensitive to the realisation at the level of 10^{-4}. The simulation discriminates sharply between kinetic models (Fig. 4b, Table 2): for the same Gaussian momentum distribution, the classical Vlasov, non-relativistic quantum and relativistic classical theories predict damping rates between 0.79 and 5.5 times the tree-level scalar-QED value, whereas the lattice result reproduces the latter. Relativistic corrections (reduced resonant velocity spread) and quantum recoil (resonance at the finite difference $g(p+k) - g(p)$) act in opposite directions and largely cancel, which is why neither correction alone is adequate.

Energy conservation, the primary diagnostic, closes the budget. (E1) The total energy of every run is conserved to 4.9×10^{-10} (E1 criterion 10^{-6}). (E2) In the cold control the fitted damping rates are at most $3.7 \times 10^{-6}\, m$ (criterion $10^{-4} m$) and the frequencies match the cold roots of Eq. (23) of Ref. [26] to 0.07%: there is no numerical dissipation that could masquerade as Landau damping. (E3) In the single-wavenumber run the coherent field energy W_A decays at $0.0570\, m$ (twice the predicted damping rate is $0.0582\, m$), and the matter energy W_ϕ absorbs it: $|\Delta(W_A + W_\phi)| / \max W_A = 4.9 \times 10^{-3}$, inside the pre-stated 10^{-2} (Fig. 4c).

4.5 Magnetar-strength fields: the spin-zero cyclotron line and its Pauli repair

Figure 5a shows the spectrum of the uniform current after the kick at $B = B_c$. With the original scheme the dominant line sits at $\omega_c = 0.58450\, m$ ($m\Delta x = 0.03125$), on the spin-zero prediction $\sqrt{m^2 + 3qB} - \sqrt{m^2 + qB} = 0.5858m$, and a weak second line marks the pair channel $E_0 + E_1$; the rest energy of the lowest Landau level is $E_0 = 1.41368\, m$ instead of m. Along the resolution ladder ($m\Delta x = 0.125$ to 0.015625) the cyclotron line deviates from the spin-$\frac{1}{2}$ value by 20.2%, 20.2%, 20.2% and 20.2% and the rest energy by 41.4%–41.4%, while the line agrees with the spin-zero prediction to 0.23%, 0.22%, 0.22% and 0.22% (Fig. 8e). The pre-stated failure criterion is met at every resolution, and the agreement with Eq. (9) certifies that the scheme solves spin-zero

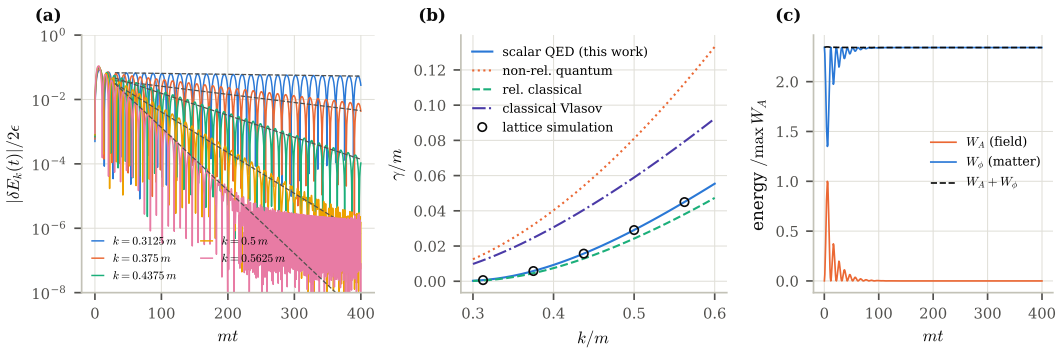


Figure 4: Landau damping with quantum recoil ($\omega_p = 0.2m$, $\sigma_p = 0.25m$). (a) Odd response $|\delta E_k(t)|/2\epsilon$ for the five wavenumbers at $N = 12288$, with the predicted decay rates (dashed). (b) Damping rate versus k: tree-level scalar-QED prediction, three reduced kinetic models, and the lattice simulation. (c) Energy budget of the perturbation for $k = 0.5m$: coherent field energy W_A, matter energy W_ϕ and their sum.

electrodynamics correctly: the failure is a property of the model, not of the discretisation. Refining the transverse cell ($L = 80/m$ and $160/m$) moves the line only to 0.58553 and $0.58579\,m$, and four realisations with different random species phases give identical lines, $\omega_c = 0.584492 \pm 7.9 \times 10^{-17}$, as they must, since species do not interfere. The failure scales as predicted with field strength (Fig. 5b): the deviation from the Dirac line is 3.4%, 10.2%, 20.2% and 29.5% for $qB/m^2 = 0.1, 0.3, 1, 3$, tracking the theoretical spin-zero/spin-$\frac{1}{2}$ differences 4.3%, 10.1%, 20.0% and 29.4% (at $qB = 0.1m^2$ the measured value is reduced by the collective shift discussed below).

Why the failure is fundamental. A spin-zero field has no magnetic moment, so the zero-point cyclotron energy $\frac{1}{2}\hbar\omega_c$ of the lowest Landau level is not cancelled by the spin energy $-\mu_B B$ as it is for an electron with $g = 2$. In the relativistic Landau spectrum this zero-point energy enters inside the square root, $E_0^2 = m^2 + |qB|$, so the lowest level of a spin-zero particle is heavier by $\sqrt{1 + B/B_c}$, and the cyclotron transition, which is the difference of two square roots, is red-shifted by a relative amount $qB/2m^2$ for $B \ll B_c$ and by 20.0% at $B = B_c$. No refinement of the lattice changes the field content of the action, so the error is independent of Δx, Δt, $\Delta_\perp$ and the ensemble. The assumption that fails is the premise, explicit in Ref. [26], that particle spin is irrelevant to plasma dynamics; it is violated as soon as $\hbar\omega_c/mc^2 \gtrsim 2\epsilon_{\rm tol}$, i.e. above $\approx 2\%$ of B_c for a 1% accuracy target.

Repair. With the Pauli coupling of Eq. (12) ($g = 2$) the dominant line moves to $\omega_c = 0.73068\,m$ and the rest energy to $0.99921\,m$ (Fig. 5a). The deviations from the spin-$\frac{1}{2}$ predictions are 0.22%, 0.19%, 0.19% and 0.19% for the line and 0.11%, 0.09%, 0.08% and 0.08% for the rest energy along the resolution ladder, within the pre-stated 1% at every resolution; transverse refinement gives 0.73173 and $0.73201\,m$, and the field scan follows the Dirac curve to 0.8%, 0.1%, 0.19% and 0.1% for $qB/m^2 = 0.1, 0.3, 1, 3$. The larger residual at $qB = 0.1m^2$, shared by the spin-zero scheme (0.9% from its own prediction), is the collective upper-hybrid shift: ω_p was held at $0.01m$ while ω_c fell to $0.09m$, so $(\omega_p/\omega_c)^2 \approx 1\%$ there. Charge is conserved to 3.2×10^{-14} and energy to 3.2×10^{-10} in the Pauli-coupled runs, the same levels as in the original scheme (1.7×10^{-14}, 3.1×10^{-10}), as anticipated from the invariance of S_P.

4.6 Spontaneous pair creation: the empty classical vacuum and its half-quantum repair

Under the Sauter pulse ($qE_0 = m^2$, $\tau = 4/m$) the predicted pair density is $n = 2.4433 \times 10^{-2}\,m$. With the original prescription—classical vacuum plus $\sigma(A) = 10^{-4}m$ link noise—the matter field remains identically zero and the measured pair density is 0, 0, 0 and 0 at $m\Delta x = 0.1, 0.05, 0.025, 0.0125$: the pre-stated failure criterion ($\leq 10^{-6}$ of the prediction) is met at every resolution, trivially and exactly. Classical noise does not cure this. Ensembles seeded with occupation ϵ per mode give pair densities $4.87 \times 10^{-4} \pm 1.6 \times 10^{-5}$ ($\epsilon = 0.01$) and $4.82 \times 10^{-3} \pm 1.6 \times 10^{-4}$ ($\epsilon = 0.1$), i.e. 1.00 and 0.99 times the stimulated prediction $2\epsilon n$ (Fig. 6a,b): the classical field creates pairs only in proportion to whatever seed it is given, and the spontaneous yield, the $\epsilon \to 0$ limit, vanishes.

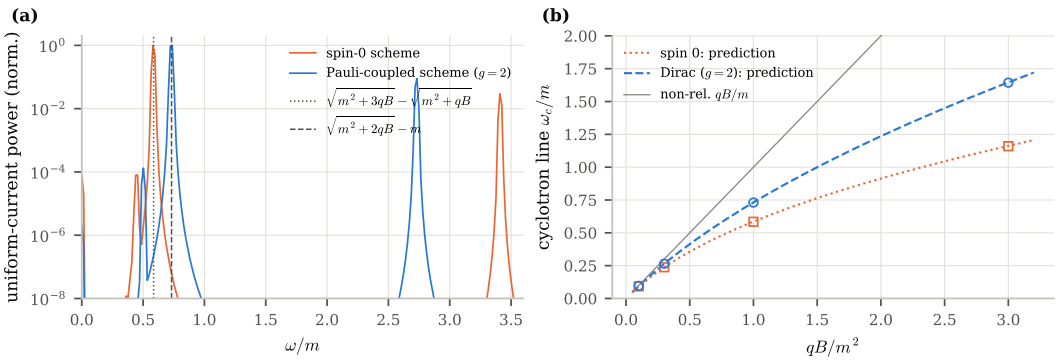


Figure 5: Cyclotron resonance of a dilute plasma in a magnetar-strength field. (a) Power spectrum of the uniform current after a small kick at $B = B_c$ ($m\Delta x = 0.03125$): original spin-zero scheme and Pauli-coupled scheme, with the spin-zero and Dirac ($g = 2$) cyclotron frequencies (vertical lines). (b) Cyclotron line versus field strength: predictions for spin zero and for spin $\frac{1}{2}$ with $g = 2$, non-relativistic qB/m, and lattice results (squares: spin-zero scheme; circles: Pauli-coupled scheme).

Why the failure is fundamental. The classical-statistical approximation replaces the quantum field by a classical field with a probability distribution over initial data; it is justified when every relevant mode has an occupation $n_p \gg \frac{1}{2}$ [21, 24, 25]. Pair creation from the vacuum is the opposite limit: all relevant modes have $n_p = 0$, so the dynamics is governed entirely by the zero-point fluctuations that the classical vacuum $\phi = 0$ omits. Because the field equation is linear in ϕ at fixed A, $\phi = 0$ is an exact fixed point, and a Bogoliubov transformation of a classical state with occupation n_p produces $2n_p|\beta_p|^2$ pairs, whereas the quantum vacuum produces $(2n_p+1)|\beta_p|^2$: the classical scheme captures exactly the stimulated part and misses exactly the spontaneous "+1". No change of resolution, gauge, or ensemble size can supply it.

Repair. With the half-quantum extension of Eq. (13) the spectrum of created pairs reproduces Eq. (11) point by point (Fig. 6a). The relative deviation of the pair density is 5.0×10^{-5}, 1.2×10^{-5}, 3.1×10^{-6} and 7.6×10^{-7} along the resolution ladder, and the largest per-mode deviation, relative to $\max_p |\beta_p|^2$, is 9.5×10^{-5}–1.5×10^{-6}, inside the pre-stated 1%; at $qE_0 = 0.5m^2$ and $2m^2$, where n changes by two orders of magnitude, the deviations are 4.6×10^{-5} and 5.4×10^{-5}. The Wigner-sampled single-field ensemble ($\epsilon = \frac{1}{2}$, 1000 members) gives $2.36 \times 10^{-2} \pm 5.1 \times 10^{-4}$, i.e. 0.97 times the prediction, confirming the equivalence of the two forms of the extension. The invariants of Sec. 2.3.2 are verified in the logs: each flavour's charge is conserved to 7.1×10^{-13}, the Bogoliubov normalisation $|\alpha_k|^2 - |\beta_k|^2 = 1$ holds to 7.0×10^{-13}, and the weighted total charge stays at 1.2×10^{-12} (absolute) while pairs are created.

Back-reaction. When the uniform field is dynamical, the original scheme leaves a supercritical field ($qE_0 = 2m^2$) untouched forever ($\min \bar{E}/E_0 = 1.000$), whereas the half-quantum extension converts it into vacuum pairs and the field undergoes plasma oscillations, reaching $\min \bar{E}/E_0 = -0.714$ (Fig. 6c), as in the semiclassical back-reaction studies of Refs. [10, 49]. The total charge stays at 9.2×10^{-14} (absolute), the Gauss residual at 8.0×10^{-12}, and the energy error relative to the initial field energy is 1.9×10^{-2}, 4.9×10^{-3} and 1.2×10^{-3} for $m\Delta x = 0.1, 0.05, 0.025$: second-order convergence, as expected from the $\mathcal{O}(q\Delta^2)$ energy law of the underlying scheme.

5 Analysis

One Bogoliubov coefficient, two outcomes. The Klein test and the vacuum test are governed by the same object. For the Sauter step, $R - 1$ in Eq. (7) is the Bogoliubov coefficient $|\beta|^2$ of the static field: each incident boson stimulates the creation of $R - 1$ pairs, which is why superradiant reflection appears at tree level and is reproduced to 9.8×10^{-6}. For the Sauter pulse, $|\beta_p|^2$ in Eq. (11) is the number of pairs created from nothing, and the tree-level scheme reproduces exactly zero. Both results follow from a single rule: a classical field with occupation n_p in the modes mixed by the field produces $2n_p|\beta_p|^2$ pairs, a quantum field $(2n_p + 1)|\beta_p|^2$. The relative error of the tree-level scheme for pair creation is therefore $1/(2n_p + 1)$—negligible for a macroscopic

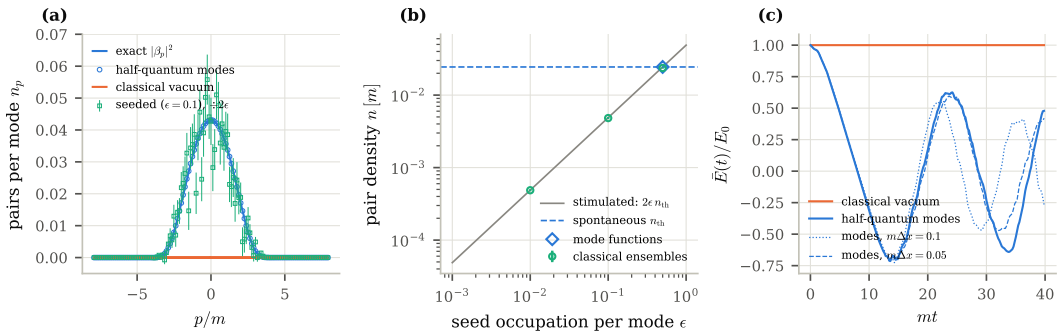


Figure 6: Spontaneous pair creation. (a) Pairs per mode created by a Sauter pulse ($qE_0 = m^2$, $\tau = 4/m$): exact $|\beta_p|^2$, half-quantum extension ($m\Delta x = 0.0125$), original classical vacuum (identically zero), and a classical-noise ensemble with seed $\epsilon = 0.1$ divided by 2ϵ. (b) Pair density versus seed occupation: classical ensembles follow the stimulated law $2\epsilon n$; the half-quantum extension gives the spontaneous density n. (c) Self-consistent decay of a supercritical field ($qE_0 = 2m^2$) into vacuum-pair plasma oscillations with the half-quantum extension (three resolutions) and absence of decay with the classical vacuum.

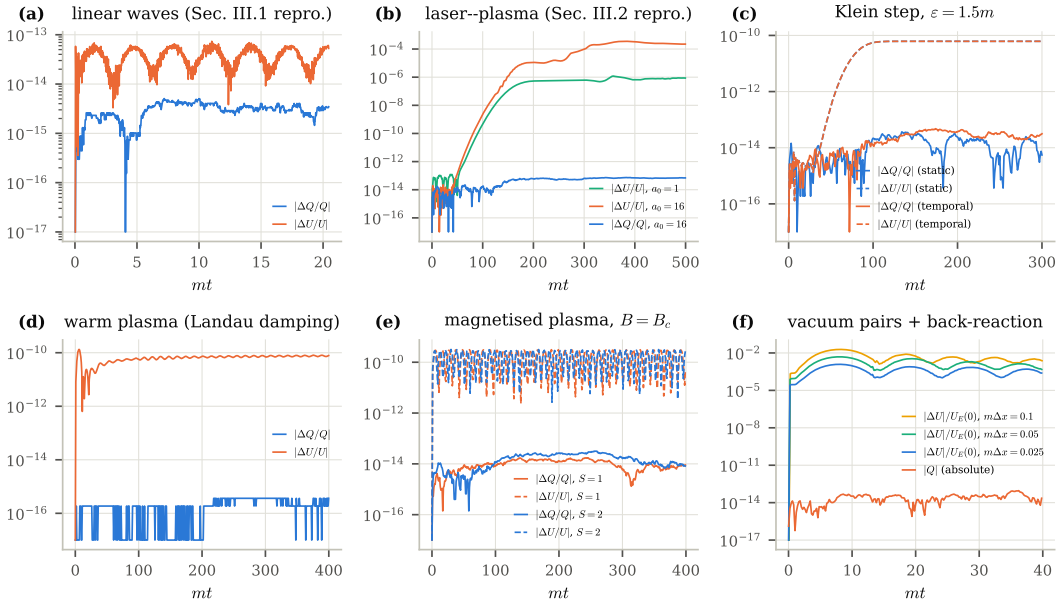


Figure 7: Conservation-law time series for one realisation of every setting: relative deviation of the total charge and of the discrete energy (for the vacuum back-reaction run the charge is zero and its absolute value is shown, and the energy error is normalised by the initial field energy).

incident wave or a dense plasma, total for the vacuum. This sharpens the validity criterion of the classical-statistical approximation for strong-field plasmas into a statement about *which modes are mixed*. The pair creation observed in the laser–plasma runs (Sec. 4.2) is stimulated by the highly occupied plasma modes and is correctly described; consistently with its Bose enhancement it sets in as soon as the laser field reaches E_c, while the wakefield is still at $0.42\,E_c$. The same field acting on weakly occupied modes—in low-density regions, or for longitudinal fields extending beyond the plasma—would produce nothing at tree level, so wherever vacuum and plasma contributions compete the half-quantum extension is required. The seeded ensembles of Sec. 4.6 measure the rule directly: their yield scales linearly with the seed, with slope $2|\beta_p|^2$.

Spin and statistics enter at different places. The Pauli extension repairs the single-particle spectrum, but the statistics remain bosonic, and the Landau-damping experiments show where

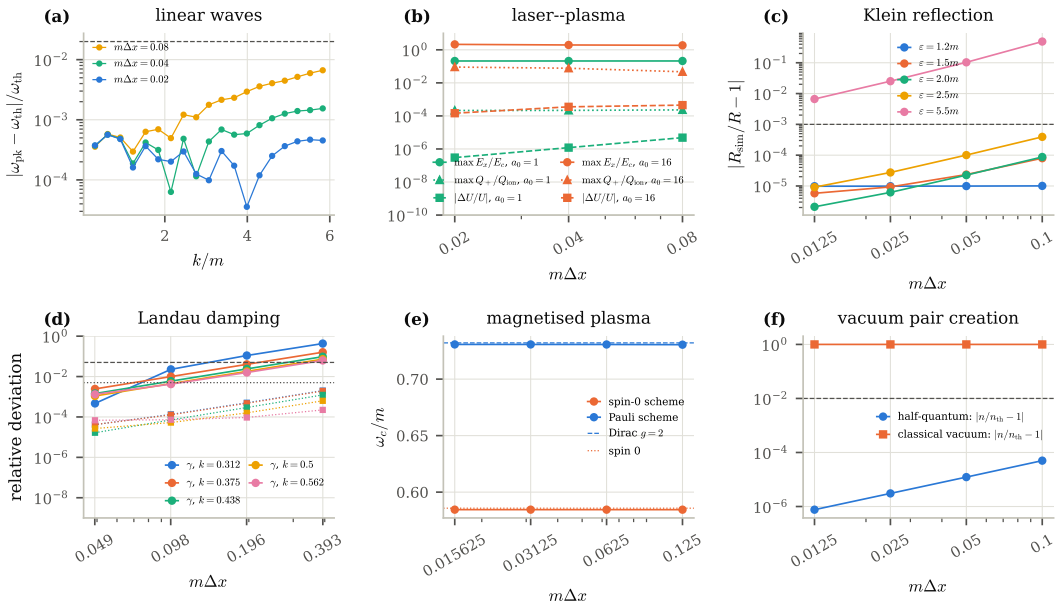


Figure 8: Convergence studies. (a) Linear waves: deviation of the transverse peak frequency from Eq. (21) of Ref. [26] at three resolutions. (b) Laser–plasma: largest wakefield, largest positron fraction and energy error for $a_0 = 1, 16$ (32 members at $m\Delta x = 0.08, 0.02$; 200 at 0.04). (c) Klein reflection: $|R_{\rm sim}/R - 1|$ (the normal-zone point $\varepsilon = 5.5m$ is shown for completeness). (d) Landau damping: deviations of γ (circles) and ω_r (squares). (e) Cyclotron line of the spin-zero and Pauli-coupled schemes. (f) Pair density from the vacuum. Dashed lines: pre-stated tolerances.

statistics re-enter. The same warm plasma represented as a *single* classical field with random phases—the textbook classical-statistical representation of a mixed state—behaves as a different physical system. A random superposition of plane waves has exponentially distributed $|\phi|^2$ (Bose bunching [84]), so the initial charge density fluctuates by order unity on the thermal de Broglie scale $1/\sigma_p$, which here exceeds the Debye length. The electrostatic energy of these fluctuations, fixed by the Gauss law at $t = 0$, is $1917\,m^3$ per transverse area, 6.9 times the entire matter energy (277, of which only 9.0 is kinetic); within $mt \approx 10$ half of it is converted into heat, raising the kinetic energy by a factor of about 97. The heated plasma then populates kinetic momenta with $p\Delta x$ of order unity, and the discrete energy, which is conserved only to $\mathcal{O}(q\Delta^2)$, drifts by 8% on average (at most 10%) over the run, whereas the same lattice conserves it to 4.9×10^{-10} in the species representation; charge remains exact. This is the field-theoretic analogue of disorder-induced heating in plasmas prepared without correlations. Consistently, the thermal $|E_k|$ exceeds the coherent signal by a factor of 192, the spread of the antithetic difference grows as $e^{0.028\,mt}$, and the ensemble-mean response of 64 realisations, measurable only up to $mt \approx 36$, oscillates at $\omega_r \approx 0.47\,m$ with $\gamma \approx 0.05\,m$ instead of the tree-level kinetic values $\omega_r = 0.31630\,m$, $\gamma = 0.02910\,m$. The number of independent modes per Debye length, $\simeq \sigma_p \lambda_D/\hbar$ up to a numerical factor, plays the role of the plasma parameter; in the quantum-recoil regime, $k \sim \sigma_p/\hbar$ with $k\lambda_D \sim 0.5$, it is necessarily of order unity, so a single bosonic field initialised with uncorrelated phases cannot represent a weakly coupled warm plasma there, however large the box. The species-resolved representation removes cross-mode interference, has no such fluctuation energy ($\mathcal{U}_A = 0$ initially), and reproduces kinetic theory; it is the classical-statistical analogue of distinguishable particles, and it is the natural representation when a scalar field models electrons, whose density fluctuations are anti-bunched rather than bunched. Spin-statistics thus affects the method twice—through the magnetic moment (repaired by S_P) and through bunching (avoided by species resolution)—and only the first is visible in the Landau spectrum.

Exact charge, approximate energy. Across all settings the total charge is conserved to between 10^{-16} and 10^{-13} (relative; absolute for the neutral vacuum), and each half-quantum flavour to 10^{-12}, independent of resolution and field strength, because it follows from an exact symmetry of S_d and of both extensions (Fig. 7). The energy, which follows from a discrete time translation that is only

approximately a symmetry of the Wilson-line coupling, is conserved to 7.4×10^{-14} for the linear waves, to 6.2×10^{-11}–3.2×10^{-10} in the Klein, Landau-damping and magnetised-plasma runs, to 1.2×10^{-6} in the weakly nonlinear laser–plasma run, to 3.5×10^{-4} when pairs are created at $a_0 = 16$, and to 1.2×10^{-3} of the field energy in the vacuum back-reaction run at the finest grid, with second-order convergence. The energy error is largest precisely when large kinetic momenta are populated (pair creation accelerates both species to $|P| \sim qE_0 t$), which identifies $P\Delta x$, rather than $qA\Delta x$, as the controlling parameter; the same parameter controls the residual resolution dependence of the $a_0 = 16$ laser–plasma run (Sec. 4.2), where the charge-weighted mean kinetic momentum reaches $6.7\,m$ ($P\Delta x \approx 0.27$) at the original resolution.

What compactness does and does not restrict. The link variables are compact, $q\Delta x A \simeq q\Delta x A + 2\pi$, and Ref. [26] therefore required $qA\Delta < 2\pi$. The Klein test shows that this is not a physical constraint: link phases that wind repeatedly change nothing, because a gauge transformation can always unwind them. What is compact is the gauge-invariant kinetic lattice momentum $P = p - qA$. A cold condensate in a sustained uniform field makes this explicit (`logs/bloch/`): instead of running away to the speed of light it Bloch-oscillates [85], with measured periods 62.83 and $125.65\,m^{-1}$ at $m\Delta x = 0.2$ and 0.1, equal to $2\pi/(qE\Delta x) = 62.83$ and $125.66\,m^{-1}$, even though its link phase winds 4.1 and 2.1 times in the same interval. The physically meaningful resolution requirement is therefore $|P|\Delta x \ll \pi$ for all populated kinetic momenta (and $\varepsilon\Delta t \ll \pi$ for all kinetic energies), a condition that a bounded potential drop always satisfies and that a sustained electromotive force always violates eventually. The latter is a genuine limitation of any fixed lattice, distinct from the two fundamental failures above in that it can be postponed indefinitely by refinement.

Where the failures begin. The two failures have simple thresholds that can be used to decide in advance whether a planned simulation needs the extensions. The spin error of the cyclotron line is $qB/2m^2$ for $B \ll B_c$; it exceeds 1% above $B \approx 0.02B_c \approx 10^8$ T, i.e. throughout magnetar magnetospheres [8, 9], and the rest-energy error $\sqrt{1 + B/B_c} - 1$ shifts every threshold that involves the lowest Landau level, including one-photon pair creation ($\omega > 2E_0$ instead of $2m$). The vacuum error is $1/(2n_p + 1)$ in the mixed modes; for a plasma of density n it is negligible when the number of quanta per mode in the volume probed by the field is large, and it is total for fields acting on vacuum. For laser fields, where $|\mathbf{E}| \sim |\mathbf{B}|$ in the lab frame, the omitted electric part of the spin coupling becomes as important as the magnetic part we restored; a full Feynman–Gell-Mann extension [83], or a lattice Dirac field with the attendant doubling problem [29], would then be needed.

Open questions. The half-quantum extension is exact for Gaussian states and external fields, but with back-reaction it is a mean-field (Hartree) theory; the Wigner-sampled single-field form contains fluctuations beyond mean field and could, in principle, describe pair–pair correlations, but it inherits the speckle problem identified above and the slow classical thermalisation of the half quanta [25]. Which of the two forms converges to the full quantum dynamics fastest in self-consistent, many-pair plasmas, and how the ultraviolet renormalisation of the half-quantum vacuum energy and current [11, 12] should be organised in three dimensions on a lattice, remain open. A second open question is whether the species-resolved representation, which removes bunching, can be combined with an exclusion constraint to model degenerate electron plasmas, where Pauli blocking rather than spin magnetism is the dominant spin-statistics effect.

6 Conclusion

Exact discrete gauge invariance makes variational lattice scalar QED a faithful solver of tree-level strong-field electrodynamics. In an independent implementation that reproduces the published linear-wave and laser–plasma demonstrations, we showed that it captures superradiant Klein reflection from a supercritical step to 9.8×10^{-6}, independent of how many times the link phases wind, and the complex frequencies of Landau-damped waves in a warm, mildly relativistic plasma with quantum recoil to within 0.24%, a regime in which reduced kinetic theories mispredict the damping rate by between 21% and a factor of 5.5. We also identified two failures that no refinement removes because they are properties of the model rather than of the discretisation: spin-zero matter misplaces Landau levels and cyclotron lines once $B \gtrsim 0.02B_c$, and the classical vacuum creates no pairs spontaneously. Both are repaired without giving up the variational structure or exact charge conservation, by a lattice Pauli coupling and by a half-quantum Bogoliubov extension of the action, respectively. The

organising principle is simple: tree-level lattice QED is exact for stimulated processes, whose rate is proportional to an occupation it represents, and blind to spontaneous ones, whose seed is the zero-point fluctuation it omits.

Limitations. All simulations are one-dimensional with a single periodic transverse cell, as in Ref. [26]; transverse dynamics enters only through prescribed transverse-momentum species, so filamentation, transverse pair creation by laser fields and the transverse phase space of Schwinger pairs are absent. The Pauli coupling restores the magnetic moment only; the electric part of the spin coupling is omitted, so the extension is exact for static magnetic fields but not for laser fields with $|\mathbf{E}| \sim |\mathbf{B}|$, and the statistics remain bosonic (no Pauli blocking). The half-quantum extension is exact for external fields; with back-reaction it is a mean-field theory without pair–pair collisions, its cost grows with the number of modes, and in three dimensions it would require an explicit renormalisation of the vacuum current and energy that we did not address. Our Landau-damping test uses the species-resolved (mean-field) representation; the single-field classical-statistical representation with uncorrelated random phases is dominated in the quantum-recoil regime by the relaxation of its initial fluctuation energy, and a correlated (screened) initial ensemble, which we did not construct, would be needed to test the same prediction with it. In the laser–plasma reproduction the $a_0 = 16$ pair yield, energy transfer and wakefield still change by 7%–18% between $m\Delta x = 0.04$ and 0.02, so only the qualitative transition, not its quantitative yields, is reproduced at the original resolution. Finally, our laser pulse follows the formula stated in Ref. [26], whose energy normalisation appears to differ from that of the published energy traces by a factor of three to four, so absolute energy-conversion fractions in the laser–plasma reproduction should not be compared one-to-one.

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